Cointegration-Based Spread Prediction for Mean-Variance Portfolio Optimisation

Evaluating whether spread prediction between cointegrated asset pairs produces more reliable return estimates than historical means for Markowitz optimisation

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Disclaimer: This report is submitted as part requirement for the BSc Degree in Computer Science at UCL. It is substantially the result of my own work except where explicitly indicated in the text. The report may be freely copied and distributed provided the source is explicitly acknowledged.

Abstract

Summarise your report concisely.

Contents

Chapter 1

Introduction

This is just a bare minimum to get started. There is unlimited guidance on using latex, e.g., `https://en.wikibooks.org/wiki/LaTeX`. You are still responsible to check the detailed requirements of a project, including formatting instructions, see `https://moodle.ucl.ac.uk/pluginfile.php/3591429/mod_resource/content/7/UGProjects2017.pdf`. Leave at least a line of white space when you want to start a new paragraph.

1.1 Section 1

Chapters should contain numbered sections and sub-sections.

1.1.1 Sub-Section 1

Remember that this is a structured technical/academic report, not an essay.

1.2 Mathematical Notation

Mathematical expressions are placed inline between dollar signs, e.g. $\sqrt{2}, \sum_{i=0}^{n} f(i)$, or in display mode

$$e^{i\pi} = -1$$

and another way, this time with labels,

$$A = B \wedge B = C \rightarrow A = C \tag{1.1}$$

$$\rightarrow C = A \tag{1.2}$$

note that

$$n! = \prod_{1 \leq i \leq n} i \tag{1.3}$$

$$\int_{x=1}^{y} \frac{1}{x} \mathrm{d}x = \log y \tag{1.4}$$

We can refer to labels like this (**??**).

Chapter 2

Background and Literature Review

The main chapters should start with a brief summary – this chapter is about, or introduces, some aspect of the work.

2.1 Main Content

Often lots of citations here (and elsewhere), e.g. [**?**] or [**?**, Theorem 2.3]. Bibtex can help with this, but is not essential. If you want pictures, try

You can use

- lists
- like this

or numbered

1. like this,
2. or this

but don't overdo it.

2.2 More Content

2.3 Summary

The main chapters should have a short summary, highlighting what the chapter has covered and any key points that need emphasising.

Chapter 3

Analysis

If you have a formal theorem you might try this.

DEFINITION 1 *See definition* **??**.

THEOREM 2 *For all* $n \in \mathbb{N},\ 1^n = 1$.

PROOF:

By induction over n. $\square$

3.1 Methodology

3.2 Implementation Details

Chapter 4

Testing and Results

4.1 Testing Methodology

4.2 Results

4.2.1 Cointegration Screening

Table 4.1: Engle-Granger Cointegration Screening Results (9 Candidate Pairs, In-Sample: Jan 2019 – Dec 2023)

Dependent	Independent	Hedge Ratio ($\hat{\beta}$)	Intercept ($\hat{\alpha}$)	ADF Stat	p-value	y is $I(1)$	x is $I(1)$	Cointegrated
GS.N	MS.N	0.780	2.351	−3.667	0.020	Yes	Yes	**Yes**
KO.N	PEP.O	0.641	0.789	−3.470	0.035	Yes	Yes	**Yes**
CVX.N	XOM.N	0.698	1.813	−3.234	0.064	Yes	Yes	No
AMD.O	NVDA.O	0.811	1.999	−2.310	0.368	Yes	Yes	No
AMZN.O	MSFT.O	0.516	2.027	−2.141	0.455	Yes	Yes	No
JNJ.N	PFE.N	0.423	3.466	−1.908	0.576	Yes	Yes	No
META.O	GOOGL.O	0.543	2.953	−1.681	0.685	Yes	Yes	No
TSM.N	NVDA.O	0.476	3.071	−1.340	0.817	Yes	Yes	No
BAC.N	JPM.N	0.854	−0.666	−1.230	0.850	Yes	Yes	No

Augmented Engle-Granger test with MacKinnon (1994) critical values at 5% significance ($p < 0.05$). Log prices used throughout.
$I(1)$ verified by ADF on levels (non-stationary) and first differences (stationary). Sorted by p-value.
Source: Author's calculations on `01_spread_exploration.ipynb`.

4.2.2 Spread Characterisation

Based on the results in Table **??**, two pairs were confirmed cointegrated at the 5% significance level: GS.N/MS.N and KO.N/PEP.O. CVX.N/XOM.N was borderline ($p = 0.064$) and excluded. Spreads for each confirmed pair are computed as the OLS residual $S_t = \ln P_t^y - \hat{\beta} \ln P_t^x - \hat{\alpha}$ and characterised by their half-life, Hurst exponent, z-score, and spread statistics.

Table **??** and Figures **??**–**??** present the spread characterisation results for the two cointegrated pairs.

Table 4.2: Spread Characterisation Summary — Cointegrated Pairs (In-Sample: Jan 2019 – Dec 2023)

Dependent	Independent	Half-Life (days)	Hurst Exponent	Current z-Score	Spread Mean (μ_S)	Spread Std (σ_S)	ADF Stat	ADF p-value
GS.N	MS.N	41.8	0.403	0.675	0.000	0.056	-3.666	0.005
KO.N	PEP.O	40.3	0.421	0.809	0.000	0.047	-3.469	0.009

Half-life estimated via AR(1)/OU regression: $\kappa = \ln(2)/\text{half-life}$ gives $\kappa \approx 0.0166$ (GS/MS) and 0.0172 (KO/PEP).
Hurst exponent $H < 0.5$ indicates mean-reverting (anti-persistent) dynamics. z-score uses 60-day rolling window at end of training period.
Spread mean is zero by construction (demeaned OLS residuals). Source: Author's calculations on `01_spread_exploration.ipynb`.

`figures/01_normalised_prices.pdf`

Figure 4.1: Normalised log price series for the two cointegrated pairs over the in-sample period (January 2019 – December 2023). Each series is normalised to zero at the start of the sample period to highlight co-movement. (a) KO/PEP (Consumer Staples); (b) GS/MS (Financials).

figures/01_spread_characterisation.pdf

Figure 4.2: Spread characterisation summary for the two cointegrated pairs (January 2019 – December 2023). (a) Half-life of mean reversion in trading days; dashed line marks the 60-day threshold. (b) Hurst exponent; dashed line marks the random walk boundary of 0.5. (c) Current z-score at the end of the training window; dotted lines mark $\pm 2\sigma$. (d) ADF p-value on the spread; dashed line marks the 5% significance level. Hatched bars indicate pairs that fail the respective threshold.

figures/01_spread_levels.pdf

Figure 4.3: Spread levels with $\pm 1\sigma$ and $\pm 2\sigma$ bands for each cointegrated pair over the in-sample period (January 2019 – December 2023). The spread is defined as $S_t = \ln P_t^y - \hat{\beta} \ln P_t^x - \hat{\alpha}$. (a) KO/PEP; (b) GS/MS.

figures/01_spread_zscores.pdf

Figure 4.4: Rolling z-scores (60-day window) for the two cointegrated pairs over the in-sample period (January 2019 – December 2023). Dotted lines mark the $\pm 2\sigma$ entry thresholds. The z-score at the end of the training window feeds directly into the OU-implied return estimation. (a) KO/PEP; (b) GS/MS.

figures/01_spread_distributions.pdf

Figure 4.5: Empirical spread distributions with overlaid normal density fits for the two cointegrated pairs (January 2019 – December 2023). The spread mean is zero by construction as the OLS residual includes an intercept. (a) KO/PEP; (b) GS/MS.

figures/01_spread_rolling_halflife.pdf

Figure 4.6: Rolling half-life of mean reversion (252-day window, EWM 21-day smoothed). Dashed line denotes the 60-day tradeability threshold. Shaded regions indicate periods where the smoothed half-life exceeded the threshold. (a) KO/PEP; (b) GS/MS.

figures/01_spread_rolling_zscore.pdf

Figure 4.7: Rolling 60-day z-score for each cointegrated pair over the in-sample period (January 2019 – December 2023). Dashed and dotted lines mark the $\pm 1\sigma$ and $\pm 2\sigma$ thresholds respectively; shaded regions indicate periods where the z-score exceeds $\pm 2\sigma$, corresponding to potential entry signals. (a) KO/PEP; (b) GS/MS.

4.2.3 Cointegration Validation

Out-of-sample spreads are constructed using hedge ratios and intercepts estimated solely on the in-sample period, ensuring no forward-looking information is incorporated. Table **??** presents the distributional shift in each spread across the IS and OOS periods, while Table **??** reports the formal statistical test results. Figure **??** presents the full IS vs OOS Engle-Granger p-value comparison across all nine candidate pairs.

Table 4.3: Spread statistics across in-sample (IS: January 2019 – December 2023) and out-of-sample (OOS: January 2024 – December 2025) periods, constructed using in-sample hedge ratios.

	Mean		Std Dev		Min		Max	
Pair	IS	OOS	IS	OOS	IS	OOS	IS	OOS
GS/MS	0.000	+0.250	0.056	0.082	−0.145	+0.067	+0.110	+0.411
KO/PEP	0.000	+0.170	0.047	0.110	−0.173	−0.016	+0.111	+0.367

GS/MS OOS minimum of +0.067 lies entirely above the IS mean, indicating the spread never revisited its historical equilibrium.
Source: Author's calculations on `02_cointegration_validation.ipynb`.

Table 4.4: Cointegration validation statistics across IS (January 2019 – December 2023) and OOS (January 2024 – December 2025) periods. ADF p-values below 0.05 indicate spread stationarity at the 5% significance level. Half-life below 60 trading days and Hurst exponent below 0.5 indicate mean reversion.

	ADF p-value		Half-Life (days)		Hurst Exponent		OOS
Pair	IS	OOS	IS	OOS	IS	OOS	z-score
GS/MS	0.005	0.452	41.8	81.1	0.403	0.362	1.015
KO/PEP	0.009	0.622	40.3	156.4	0.421	0.575	0.604

Neither spread remains stationary OOS using IS hedge ratios. KO/PEP Hurst crosses above 0.5, indicating transition to near-random-walk behaviour.
Source: Author's calculations on `02_cointegration_validation.ipynb`.

figures/02_oos_eg_pvalues.pdf

Figure 4.8: IS vs OOS Engle-Granger p-values for all nine candidate pairs. Hatched bars indicate pairs that fail the 5% significance threshold. GS/MS is the only pair persistent across both periods. KO/PEP degrades OOS; NVDA/TSM appears cointegrated OOS but is invalidated by NVDA.O failing the $I(1)$ precondition due to explosive price growth in 2024–2025.

figures/02_spread_is_oos_concat.pdf

Figure 4.9: Spread levels across the full sample period (January 2019 – December 2025). The dashed vertical line marks the IS/OOS boundary (January 2024). Horizontal lines and shading denote IS $\pm 1\sigma$ and $\pm 2\sigma$ bands; persistent OOS displacement beyond these bounds indicates a structural break in the cointegrating relationship. (a) KO/PEP; (b) GS/MS.

figures/02_spread_oos_levels.pdf

Figure 4.10: Spread levels over the out-of-sample period (January 2024 – December 2025), constructed using in-sample hedge ratios and intercepts. Horizontal lines denote the IS mean and $\pm 2\sigma$ bands estimated over the training window. (a) KO/PEP; (b) GS/MS.

figures/02_zscore_is_oos.pdf

Figure 4.11: Rolling 60-day z-score across the full sample period (January 2019 – December 2025). The dashed vertical line marks the IS/OOS boundary. Shaded regions indicate $|z| > 2\sigma$ exceedances. Persistent non-zero z-scores in the OOS period are consistent with a shift in the spread equilibrium level. (a) KO/PEP; (b) GS/MS.

figures/02_rolling_halflife_is_oos.pdf

Figure 4.12: Rolling half-life of mean reversion across the full sample period (January 2019 – December 2025). Faint lines show raw estimates; bold lines show 21-day exponentially weighted smoothed values. The dashed horizontal line marks the 60-day tradability threshold. The dashed vertical line marks the IS/OOS boundary. (a) KO/PEP; (b) GS/MS.

4.2.4 Return Estimation

OU-implied expected returns are derived from the end-of-IS z-score for each confirmed pair. A positive z-score (spread above mean) implies the spread is expected to fall, producing a negative OU-implied return; a negative z-score implies the spread is expected to rise, producing a positive return. Table **??** presents the sanity check confirming directional consistency between the z-score and OU-implied return signal.

Table 4.5: End-of-sample z-score validation and OU-implied expected returns (IS period: Jan 2019 – Dec 2023)

Pair	End-of-IS z-Score	OU-Implied μ (ann.)	Historical μ (ann.)	Signal	Consistent
GS/MS	+1.739	−10.69%	GS: +11.9%, MS: +15.5%	Spread well above mean	Yes
KO/PEP	−0.759	+1.93%	KO: +6.4%, PEP: +8.3%	Spread below mean	Yes

OU-implied μ computed via rolling 60-day OU model, annualised at 252 trading days per year.
Historical μ is the annualised sample mean of daily returns over the IS period.
Source: Author's calculations on `03_return_estimation.ipynb`.

figures/03_cumulative_returns.pdf

Figure 4.13: Cumulative returns of each cointegrated spread versus its constituent assets over the in-sample period (January 2019 – December 2023). All series are indexed to 1.0 at the start. The spread return reflects the daily change in the OLS residual $S_t = \ln P_t^y - \hat{\beta} \ln P_t^x - \hat{\alpha}$. (a) KO/PEP; (b) GS/MS.

figures/03_mu_comparison.pdf

Figure 4.14: Expected return estimates (annualised, IS 2019–2023). (a) OU-implied μ by pair: GS/MS has a large negative implied return (spread well above equilibrium at end of IS); KO/PEP has a small positive implied return (spread slightly below equilibrium). (b) Historical mean μ by ticker: all four assets exhibit positive historical means, producing a very different allocation under mean-variance optimisation.

figures/03_correlation_heatmap.pdf

Figure 4.15: Asset return correlation matrix for the four constituent tickers (GS.N, KO.N, MS.N, PEP.O) estimated over the in-sample period (January 2019 – December 2023) using Ledoit-Wolf shrinkage covariance. GS and MS exhibit the highest within-pair correlation, consistent with their financial sector co-movement.

4.2.5 Backtesting Results

Table **??** reports out-of-sample portfolio performance metrics over the test period (January 2024 – December 2025). The OU-Implied MPT portfolio operates in spread space (GS/MS and KO/PEP as assets) under minimum-variance weights; the Historical MPT portfolio operates in asset space (GS.N, KO.N, MS.N, PEP.O) under minimum-variance weights with historical mean returns.

Table 4.6: Out-of-sample portfolio performance metrics (test period: January 2024 – December 2025). Vol. Reduction is relative to the equal-weight baseline; negative values indicate higher volatility than equal-weight.

	Sharpe Ratio	Max Drawdown	Total Return	Ann. Volatility	Vol. Reduction
OU-Implied MPT	1.359	−8.7%	+34.2%	9.8%	+0.5%
Historical MPT	0.335	−14.3%	+12.1%	14.1%	−43.4%
Equal-Weight	1.363	−8.4%	+34.5%	9.8%	—
Buy & Hold	1.403	−30.0%	+108.0%	27.5%	—
S&P 500 (SPY)	1.110	−18.9%	+44.3%	16.0%	—

Minimum-variance weights used for both MPT portfolios. Risk-free rate: 2% p.a.

Source: Author's calculations on `04_backtest_results.ipynb`.

Table 4.7: Per-pair z-score mean-reversion backtest metrics (OOS: January 2024 – December 2025). Entry at $\pm 2.0\sigma$, exit at $\pm 0.5\sigma$.

Pair	Sharpe	Max DD	Total Return	Ann. Vol	Trades	Days in Position	Win Rate
GS.N/MS.N	−0.285	−20.7%	−8.4%	17.2%	9	283	50.2%
KO.N/PEP.O	−0.665	−15.1%	−6.3%	7.5%	8	312	47.8%

Negative Sharpe ratios are consistent with OOS cointegration breakdown identified in `02_cointegration_validation.ipynb`.

Source: Author's calculations on `04_backtest_results.ipynb`.

figures/04_cumulative_returns.pdf

Figure 4.16: Cumulative growth of $1 invested at the start of the OOS period (January 2024 – December 2025) for the OU-Implied MPT, Historical MPT, equal-weight, and S&P 500 benchmarks. The OU-Implied and equal-weight portfolios track closely throughout; the Historical MPT underperforms substantially.

figures/04_drawdown.pdf

Figure 4.17: Peak-to-trough drawdown over the OOS period (January 2024 – December 2025). Lower (closer to zero) is better. The OU-Implied MPT achieves a maximum drawdown of -8.7%, substantially lower than the Historical MPT (-14.3%) and S&P 500 (-18.9%).

figures/04_rolling_sharpe.pdf

Figure 4.18: Rolling 60-day Sharpe ratio over the OOS period (January 2024 – December 2025). Time-varying risk-adjusted performance computed over a trailing 60-day window. Periods above zero indicate positive risk-adjusted returns in that window.

figures/04_rolling_volatility.pdf

Figure 4.19: Rolling 60-day annualised volatility over the OOS period (January 2024 – December 2025). The OU-Implied MPT maintains substantially lower and more stable volatility than the Historical MPT throughout the test period.

figures/04_weights_bar.pdf

Figure 4.20: Minimum-variance portfolio weights for both approaches. (a) OU-Implied MPT: 46.6% GS/MS spread, 53.4% KO/PEP spread. (b) Historical MPT: 11.7% GS.N, 48.7% KO.N, 0.0% MS.N, 39.7% PEP.O. The historical allocation concentrates heavily in KO and PEP due to their lower variance, while zeroing out MS.N.

figures/04_pair_cumulative_returns.pdf

Figure 4.21: Cumulative returns from the z-score mean-reversion backtest engine for each cointegrated pair over the OOS period (January 2024 – December 2025). Both pairs generate negative cumulative returns, consistent with persistent spread drift rather than mean reversion under IS-calibrated thresholds. (a) GS.N/MS.N; (b) KO.N/PEP.O.

figures/04_pair_zscores.pdf

Figure 4.22: Rolling z-scores with entry ($\pm 2.0\sigma$) and exit ($\pm 0.5\sigma$) thresholds for each cointegrated pair over the OOS period. Trade entry and exit points are marked. Persistent positive z-scores reflect the level shift in both spreads identified in the validation stage. (a) GS.N/MS.N; (b) KO.N/PEP.O.

figures/04_efficient_frontier.pdf

Figure 4.23: Mean-variance efficient frontiers for both the OU-Implied (spread space) and Historical (asset space) portfolios. The two frontiers are not directly comparable as they operate over different asset universes. The minimum-variance portfolios used in the backtest are marked on each frontier.

figures/04_return_distribution.pdf

Figure 4.24: Histogram of daily portfolio returns for the OU-Implied and Historical MPT portfolios over the OOS period (January 2024 – December 2025). The OU-Implied distribution is narrower and more concentrated around positive values, reflecting its lower volatility and higher mean return. (a) OU-Implied MPT; (b) Historical MPT.

Chapter 5

Dashboard

Chapter 6

Discussion

6.1 Interpretation of Results

Table **??** shows that both cointegrated pairs exhibit half-lives between 40 and 42 trading days and Hurst exponents below 0.5, confirming statistically significant mean-reverting behaviour over the in-sample period. ADF tests on the spread residuals reject the unit root hypothesis at the 1% significance level for both pairs, providing strong empirical support for the spread-based return estimation approach in-sample.

However, Table **??** reveals that neither spread remains stationary OOS when IS hedge ratios are applied, with ADF p-values rising to 0.452 and 0.622. This OOS breakdown directly affects the reliability of IS-calibrated return signals, as evidenced by the negative Sharpe ratios in the per-pair backtest (Table **??**). Despite this, the portfolio-level OU-Implied MPT substantially outperforms the Historical MPT on all three evaluation metrics (Table **??**), suggesting that the structural properties of the spread universe — low volatility and low market beta — provide resilience beyond the precision of the OU return estimates alone.

Chapter 7

Conclusions and Evaluation

7.1 Achievements

Summarise the achievements to confirm the project goals have been met.

7.2 Evaluation

Evaluation of the work (this may be in a separate chapter if there is substantial evaluation).

7.3 Future Work

How the project might be continued, but don't give the impression you ran out of time!

The project could have continued by adding a rebalance option which allows the functionality of producing a portfolio with percentage of each asset and capital allocated in a way to minimise the number of transactions needed, to minimise transactional cost.

Bibliography

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Appendix A

Selected US Equities: Candidate Universe for Pair Selection

Table A.1: US Equity Candidate Universe ranked by Market Capitalisation

Rank	Company	Symbol	Market Cap in USD (29 Dec 2023)
1	Microsoft	MSFT.O	$2,795B
2	Alphabet	GOOGL.O	$1,755B
3	Amazon	AMZN.O	$1,570B
4	Nvidia	NVDA.O	$1,223B
5	Meta	META.O	$910B
6	TSMC	TSM.N	$501B
7	JPMorgan	JPM.N	$492B
8	ExxonMobil	XOM.N	$400B
9	J&J	JNJ.N	$377B
10	Costco	COST.O	$293B
11	Chevron	CVX.N	$281B
12	BofA	BAC.N	$266B
13	Coca-Cola	KO.N	$255B
14	AMD	AMD.O	$238B
15	PepsiCo	PEP.O	$234B
16	Intel	INTC.O	$212B
17	Pfizer	PFE.N	$163B
18	Morgan Stanley	MS.N	$153B
19	Goldman Sachs	GS.N	$126B
20	Schlumberger	SLB.N	$74B
21	Target	TGT.N	$66B
22	Halliburton	HAL.N	$32B
23	Delta Airlines	DAL.N	$26B
24	United Airlines	UAL.O	$14B

Appendix B

Design Diagram of the Dashboard with Excalidraw

Appendix C

Project Plan

Appendix D

Interim Report